

Business Problem & Requirements

End-to-End Data Analytics Project Documentation

1. Executive Summary

In the highly competitive e-commerce landscape, delivery speed and reliability are critical drivers of customer retention. Currently, our global retail network is experiencing a 15% increase in customer complaints related to delayed shipments and missing packages. Concurrently, our outbound freight spend has increased by 8% quarter-over-over without a proportional increase in order volume.

2. Problem Statement

The business lacks end-to-end visibility into the supply chain lifecycle. We cannot currently identify whether delivery delays are originating internally (slow warehouse processing) or externally (poor carrier performance). Furthermore, we do not have a clear understanding of the cost-to-performance ratio across our fulfillment network.

Core Business Objectives

- **Improve On-Time Delivery (OTD):** Identify root causes of late deliveries to increase OTD rate to 95%.
- **Optimize Freight Spend:** Evaluate carrier performance to reallocate shipping volume to the most cost-effective and reliable partners.
- **Enhance Internal Efficiency:** Isolate warehouse processing times to identify specific fulfillment centers requiring operational audits.

3. Project Scope & Deliverables

- **Data Preparation (Python):** Clean historical shipping data, handle missing timestamps, and calculate accurate SLA metrics (Processing Time, Transit Time).
- **Exploratory Analysis (SQL):** Query the structured dataset to extract key performance indicators (KPIs) grouped by carrier, region, and facility.
- **Interactive Dashboard (Power BI):** Develop a tracking tool for supply chain managers to monitor real-time fulfillment metrics.

- **Strategic Recommendations:** Present actionable, data-driven insights to the executive logistics team.